

Letizia Lamperti

Phd in AI for Ecology

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Github in LinkedIn



Research interests

Biodiversity monitoring | Machine Learning | Deep Learning | Neural Networks

Education

2018–2020 : **Master of Science of Mathematical Engineering, major: Applied Statistics** *Politecnico di Milano*.

2014–2018 : **Bachelor of Mathematical Engineering** *Politecnico di Milano*.

Research experience

2025–2026 **Postdoctoral Fellow** *TIMC-IMAG, Université Grenoble Alpes, France*.

Within the project *Models for Predictive Genomic Ecology* (ANR-22-CE45-0033), this postdoctoral research develops statistical and computational methods to predict ecological community responses to environmental change. The work integrates genomic, ecological, environmental and functional data into predictive frameworks, with a particular focus on species distribution models. It also involves close collaboration with interdisciplinary teams in quantitative ecology, bioinformatics and applied mathematics, with the aim of advancing methodological tools for biodiversity forecasting and ecosystem management.

2021–2024 **PhD in Machine Learning and Deep Learning Methods in Ecology** *CEFE (Montpellier, France) – ETH Zurich (Switzerland) – TIMC-IMAG (Grenoble, France)*.

Thesis conducted within the *AI4theSciences* doctoral programme (Marie Skłodowska-Curie Actions, Horizon 2020), under the supervision of S. Manel, L. Pellissier and O. François. The research focused on the methodological development of machine learning and deep learning approaches to enhance the use of environmental DNA (eDNA) metabarcoding as a biomonitoring tool in marine ecosystems. It introduced novel computational frameworks to overcome the limitations of traditional analytical methods, addressing the complexity of high-dimensional eDNA data. The work combined analyses on both raw sequences and aggregated biodiversity indicators, enabling the detection of biodiversity variations, the identification of socio-environmental signals, and the monitoring of ecosystem dynamics. These advances contribute scalable tools for biodiversity assessment and conservation strategies in marine environments.

Summers **Fieldwork (as part of PhD)** *Western Mediterranean Sea*.

2022–2023 Participation in the *BioDivMed* biodiversity monitoring programme, with responsibilities including the collection of eDNA samples in marine ecosystems.

Jan 2021 – Jun 2021 **Research Internship** *Instituto Universitario de Oceanografía y Cambio Global (IOCAG), Gran Canaria, Spain*.

Research internship supervised by S. Hernández-León and M. Matteucci. The project focused on the development of computer vision methods for the automatic classification of phytoplankton using deep learning algorithms. These approaches were applied in the context of climate change research and Atlantic Ocean biodiversity monitoring (ATLAS).

Publications

Journal Articles and Preprints

- 2025 **Lamperti L.**, Sanchez T., Stalder S., Mouillot D., Volpi M., Albouy C., Bruno M., Molin N., François O., Manel S., & Pellissier L., End-to-End Artificial Learning of Protection Gradient from Raw eDNA Sequences. *Authorea Preprint*, September 17, 2025. DOI: 10.22541/au.175812518.84262613/v1.
- 2025 Sanchez T., Stalder S., **Lamperti L.**, Brosse S., Frossard A., Leugger F., Rozanski R., Zong S., Manel S., Medici L., Kuhn F., Han X., Mestrot A., Albouy C., Volpi M., & Pellissier L., ORDNA: Deep-learning-based ordination for raw environmental DNA samples. *Methods in Ecology and Evolution*, 2025. DOI: 10.1111/2041-210X.70033.
- 2024 **Lamperti L.**, François O., Mouillot D., Mathon L., Sanchez T., Albouy C., Pellissier L., & Manel S., A spatial matrix factorization method to characterize ecological assemblages as a mixture of unobserved sources: An application to fish eDNA surveys. *Methods in Ecology and Evolution*, 15, 2301–2315. DOI: 10.1111/2041-210X.14430.
- 2023 **Lamperti L.**, Sanchez T., Si Moussi S., Mouillot D., Albouy C., Flück B., Bruno M., Valentini A., Pellissier L., & Manel S., New deep learning-based methods for visualizing ecosystem properties using environmental DNA metabarcoding data. *Molecular Ecology Resources*, 23, 1946–1958. DOI: 10.1111/1755-0998.13861.

Oral Presentations and Conferences

- 2025 Perspective IA & Biodiversité, CNRS, Paris, France.
- 2025 StaTalk 2025, University of Milano-Bicocca, Italy.
- 2024 International Conference Artificial Intelligence for the Sciences (AI4TheSciences), Paris, France. *Machine Learning for Biodiversity Monitoring*.
- 2024 GDR Ecostat, Montpellier, France. *Unveiling Ecological Assemblages: A Spatial Matrix Factorization Approach for Analyzing Fish eDNA Surveys*.
- 2023 SMBE 2023, Ravenna, Italy. *Using transformers to relate DNA sequence composition from eDNA to ecosystem properties*.
- 2023 Environmental DNA Hobart 2023, Hobart, Australia (virtual). *Deep Learning Methods for eDNA*.
- 2023 & 2022 PRAIRIE Seminars, Grenoble and Paris, France.

Teaching experience

- 2025 Statistics UE STA301, 20h, *Université Grenoble Alpes* Grenoble, France.
- 2025 Intelligent Systems – Introduction to AI, 6h, *ENSIMAG* Grenoble, France.
- 2024 Deep Learning for eDNA, 2h, *BioEcoTrop*, *University of La Réunion* St Denis, France.
- 2023 Biodiversity monitoring – Environmental DNA, 6h, *EPHE – Conservation Sciences* Montpellier, France.

Visiting

- May 2025 IFREMER – Délégation Océan Indien La Réunion, France.
- Apr 2024 IFREMER – Délégation Océan Indien La Réunion, France.

Skills

- Software Advanced user and developer: **R** and **Python**.
Intermediate user: **Matlab**, **C++**, **Latex**.
Proficient in the use of high-performance computing (HPC) clusters for AI.

Languages Italian (mother tongue), French (written and spoken), English (written and spoken), Spanish (written and spoken).

Services

Reviewer Molecular Ecology, Environmental DNA, Ecology Letters.

Center of interest

Passionate about sports and adventures, including: Rock climbing, Traveling, Sailing and Surfing.